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MRS-SHAP: Multi-objective explainable EEG feature selection framework for ADHD detection

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Abstract

Attention Deficit Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder characterized by impairments in attention, impulsivity, and executive function. Current diagnostic approaches rely primarily on subjective behavioral assessments, highlighting the need for objective and interpretable neurophysiological biomarkers. This study proposes a Multi-Objective Rank Scored–SHapley Additive exPlanations (MRS-SHAP) framework for EEG-based ADHD analysis, integrating explainability, temporal stability, and performance-aware feature selection within a unified pipeline. A publicly available dataset comprising 121 children (61 ADHD, 60 controls; age 7–12 years), recorded using a 19-channel EEG system, was utilized. A total of 998 features were extracted across spectral, nonlinear, and connectivity domains and dataset does not include subtype-specific ADHD labels; therefore, the analysis focuses on general ADHD classification. The proposed approach combines SHAP-based feature importance, Dynamic Time Warping (DTW)-based temporal stability, and cross-validated performance weighting to derive a compact, robust, and explainable feature subset. All preprocessing, resampling (SMOTE-ENN), feature selection, and model training steps were implemented within a strictly nested, leakage-free cross-validation framework to ensure unbiased evaluation. Under segment-level evaluation, the CatBoost classifier achieved high performance (Accuracy = 98.40%, F1-score = 98.40%), representing an upper-bound estimate. However, subject-level validation using GroupKFold yielded more realistic performance (Accuracy ≈ 79.5%, AUC ≈ 0.86), while evaluation on a held-out test set (~20% of subjects) produced a more conservative estimate (Accuracy ≈ 65.7%, AUC ≈ 0.73). This performance gap highlights the impact of inter-subject variability and emphasizes the importance of strict subject-level validation in EEG-based modeling. Comparative analysis demonstrates that the proposed feature selection approach outperforms conventional methods such as PCA, ANOVA, and Recursive Feature Elimination (RFE). SHAP-based interpretability identifies beta-band activity, EEG microstate dynamics, and fronto-central connectivity as key discriminative features, consistent with established neurophysiological evidence of ADHD-related network dysfunction. Ablation and sensitivity analyses further confirm the complementary contributions of feature importance and temporal stability, with minimal dependence on synthetic resampling. These findings indicate that integrating explainability and temporal



stability enhances the robustness and interpretability of EEG-based feature selection. However, the results should be interpreted as methodological validation rather than clinical evidence, as the study is limited to a single dataset and may be influenced by confounding factors such as medication exposure. Future work will focus on external validation across independent cohorts and clinically controlled settings to assess generalizability and translational relevance.

Keywords ADHD prediction, EEG Signal Analysis, Multi-Objective Feature Selection, Explainable Machine Learning, MRS-SHAP, EEG Microstates

1 Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a widespread neurodevelopmental disorder which arises from the presence of inattention, impulsiveness, and erratic mind-space, and if displayed to a great degree, it represents the cardinal symptoms of ADHD and slowly leads to functional impairment. ADHD is a widespread neurodevelopmental disorder which arises from the presence of inattention, impulsiveness, and erratic mind-space, and if displayed to a great degree, it represents the cardinal symptoms of ADHD and slowly leads to functional impairment [1]. Even though ADHD is most diagnosed in children, early recognition and intervention remain limited in many parts of the world due to low awareness and limited access to therapeutic services [2]. The condition is notably more prevalent in boys than in girls, with a global prevalence rate of approximately 8% among boys, reflecting a male-to-female ratio of 2–3:1 [3]. Many people with ADHD continue to experience symptoms well into adulthood, with studies showing that it affects around 2.5–3.1% of adults. This condition can lead to struggles with mental health, such as difficulties managing emotions, impulsive or disruptive behaviors, and even substance use [3, 4].

ADHD is mainly recognized as a part of a spectrum disorder comprising three primary subtypes: hyperactive/impulsive (ADHD-HI), inattentive (ADHD-IA), and combined type (ADHD-C). Although ADHD-C and ADHD-IA both involve attention deficits, they exhibit distinct cognitive and neurological profiles. Children with ADHD-C often display poor impulse control, impacting working memory, sustained attention, and response inhibition, and are frequently affected by comorbid conditions such as sleep disturbances and anxiety [5, 6]. On the contrary, ADHD-IA is associated with vigilance impairments and increased social anxiety, with neuroimaging studies indicating increased gray matter volume in the left precentral gyrus and posterior cingulate cortex [7]. Furthermore, individuals with ADHD-IA show reduced gray matter in the left dorsolateral prefrontal cortex, a region integral to impulse control, compared to those with ADHD-C [8, 9]. Despite these findings, current diagnostic frameworks fail to include such neurostructural distinctions. Developmentally, ADHD also demonstrates age-related brain volume changes, with younger patients (≤ 15 years) suggesting smaller caudate volumes, while older individuals (> 22 years) tend to exhibit larger caudate structures [10, 11].

Conventional methods for detecting ADHD largely rely on behavioral observations, parent-teacher reports, and structured clinical assessments, which remain the current gold standard. Nonetheless, these approaches are susceptible to subjective bias. Another tool often used is the computer-based continuous performance task (CPT), where individuals are exposed to infrequent visual stimuli interspersed with prolonged monotonous distractions, and their response patterns are assessed. Though CPTs offer some

indication of attention-related deficits, their diagnostic accuracy for ADHD remains limited [5]. A more neutral approach involves electrophysiological measures particularly electroencephalography (EEG), which captures electrical fields generated by neuronal activity. EEG is a frequently used method in ADHD research due to its cost-effectiveness and non-invasiveness, having been employed for over four decades to identify, characterize, and quantify brainwave patterns [12]. Individuals with ADHD often exhibit elevated theta (4–8 Hz) and reduced beta (13–30 Hz) wave activity, giving a high theta/beta ratio, which has been proposed as a potential biomarker for ADHD detection [13]. Further supporting the utility of EEG-based feature analysis, recent studies have demonstrated the effectiveness of combining time and frequency domain features with machine learning models for cognitive state estimation, including applications in surgical workload assessment using random forest regression [14].

Furthermore, event-related potential (ERP) components, particularly the P300 wave, which is associated with attention and information processing, often show reduced amplitude or delayed latency in ADHD populations. The Eriksen Flanker Test has been majorly used to examine aspects of cognitive control such as flexibility, response conflict, and performance monitoring in ADHD [12]. In a study by Mazaheri et al. (2013), suppression of occipital alpha activity (8–12 Hz) was evaluated as a cue-processing index, highlighting the role of alpha oscillations in modulating attention by gating irrelevant sensory information [15]. Furthermore, cognitive stimulation paradigms involving progressive cognitive load have demonstrated the potential to normalize alpha band activity, effectively aligning EEG profiles of children with ADHD closer to those of neurotypical controls [16]. Recent advancements in EEG-based analysis have demonstrated the effectiveness of explainable AI techniques such as SHAP for interpreting model decisions. For example, a recent study in Adhikari et al. [17] demonstrated utilized variational mode decomposition (VMD)-based frequency domain features combined with SHAP to analyze emotion-related EEG patterns, achieving high classification accuracy using a random forest model. Similarly, a study in Mouazen et al. [18] employed hybrid deep learning architectures on the DEAP dataset with SHAP-based interpretability, highlighting the balance between accuracy, computational efficiency, and real-time applicability in EEG-driven emotion recognition.

Maniruzzaman et al. (2023) [19] employed a hybrid channel selection approach combining SVM-based and t-test based methods, following with LASSO logistic regression and classifiers (including GPC, RF, k-NN, MLP, decision tree) showed success in detecting ADHD using EEG signals with an accuracy of 97.53%, precision of 97% and sensitivity of 98.46%. In the study done by Maniruzzaman et al. (2022) [20], using t-test and LASSO methods and four classifiers, namely SVM with RBF kernel, k-NN, MLP and Logistic Regression, they achieved 94.2% accuracy and 93% sensitivity. Lastly, in R. M. I. S. Kumarasingha et al.'s study (2023) [2], which focused on early detection of ADHD, used eight classifiers-RF, Naïve Bayes, Decision Tree, XGBoost, k-NN, MLP, SVM, and 1D CNN, out of which RF gave the highest performance metrics-accuracy of 91.67%, sensitivity of 95% and precision of 92%. Recent studies have explored the integration of data balancing strategies with explainable artificial intelligence in neurophysiological classification tasks. For instance, Trabassi et al. (2024) [21] demonstrated that combining generative data augmentation with interpretability techniques can enhance robustness and transparency in small and imbalanced datasets. These findings highlight the growing

importance of integrating data-level balancing with explainable modeling approaches. In contrast, the proposed method focuses on multi-objective feature selection by combining SHAP-based importance, temporal stability, and performance-aware ranking, offering an alternative strategy for improving model reliability without relying on generative augmentation.

Furthermore, emerging neurophysiological evidence suggests that ADHD is characterized by alterations in both neural complexity and large-scale brain connectivity. Aydın et al. (2022) [22] demonstrated that children with ADHD exhibit disrupted functional brain networks, including reduced baseline connectivity and altered global network indices, supporting the view of ADHD as a network-level disorder rather than a purely localized dysfunction. Complementing this, Çetin et al. (2022) [23] reported that neuro-cortical complexity, quantified using entropy-based measures, is significantly altered in children with ADHD and can be modulated by stimulant treatment. These findings indicate that ADHD involves both abnormal signal irregularity at the local level and impaired coordination across distributed brain regions. Together, these studies suggest that ADHD is governed by a dual mechanism involving altered regional neural complexity and dysfunctional inter-regional communication. This provides a strong rationale for integrating both single-channel analyses, which capture local spectral and nonlinear dynamics, and multi-channel approaches, which characterize functional connectivity and large-scale network interactions. Motivated by this, the present study adopts a multi-domain EEG feature extraction strategy that combines spectral, nonlinear, and connectivity-based features, enabling a comprehensive representation of ADHD-related neural dysfunction.

The primary objective of this study is to develop a multi-objective, explainable feature selection framework for EEG-based ADHD analysis, rather than to propose a new classification model. Specifically, the proposed MRS-SHAP framework is designed to integrate SHAP-based feature importance, Dynamic Time Warping (DTW)-based temporal stability, and performance-aware weighting into a unified ranking strategy for identifying robust, explainable, and neurophysiologically relevant EEG features. The framework aims to capture complementary aspects of brain activity, including spectral, temporal, and connectivity-based characteristics, while ensuring that the selected features are both discriminative and stable across samples. Given that the dataset does not include subtype-specific annotations, the analysis focuses on identifying generalizable EEG patterns associated with pediatric ADHD rather than subtype differentiation. Classification performance is used as an evaluation mechanism to assess the quality of the selected feature subset, rather than as the primary contribution of the study. Accordingly, the proposed framework is positioned as a methodological advancement toward explainable and robust EEG feature selection, with potential implications for future biomarker discovery and clinical translation.

2 Methodology

The systematic processing pipeline of the proposed computational framework is illustrated in Fig. 1, which depicts the sequential transformation of raw EEG data through preprocessing, feature extraction, and machine learning classification stages. Each stage is designed to enhance the signal quality, extract neurophysiologically relevant features, and identify patterns that differentiate attention deficit hyperactivity disorder (ADHD) subjects from neurotypical controls. The proposed framework, termed

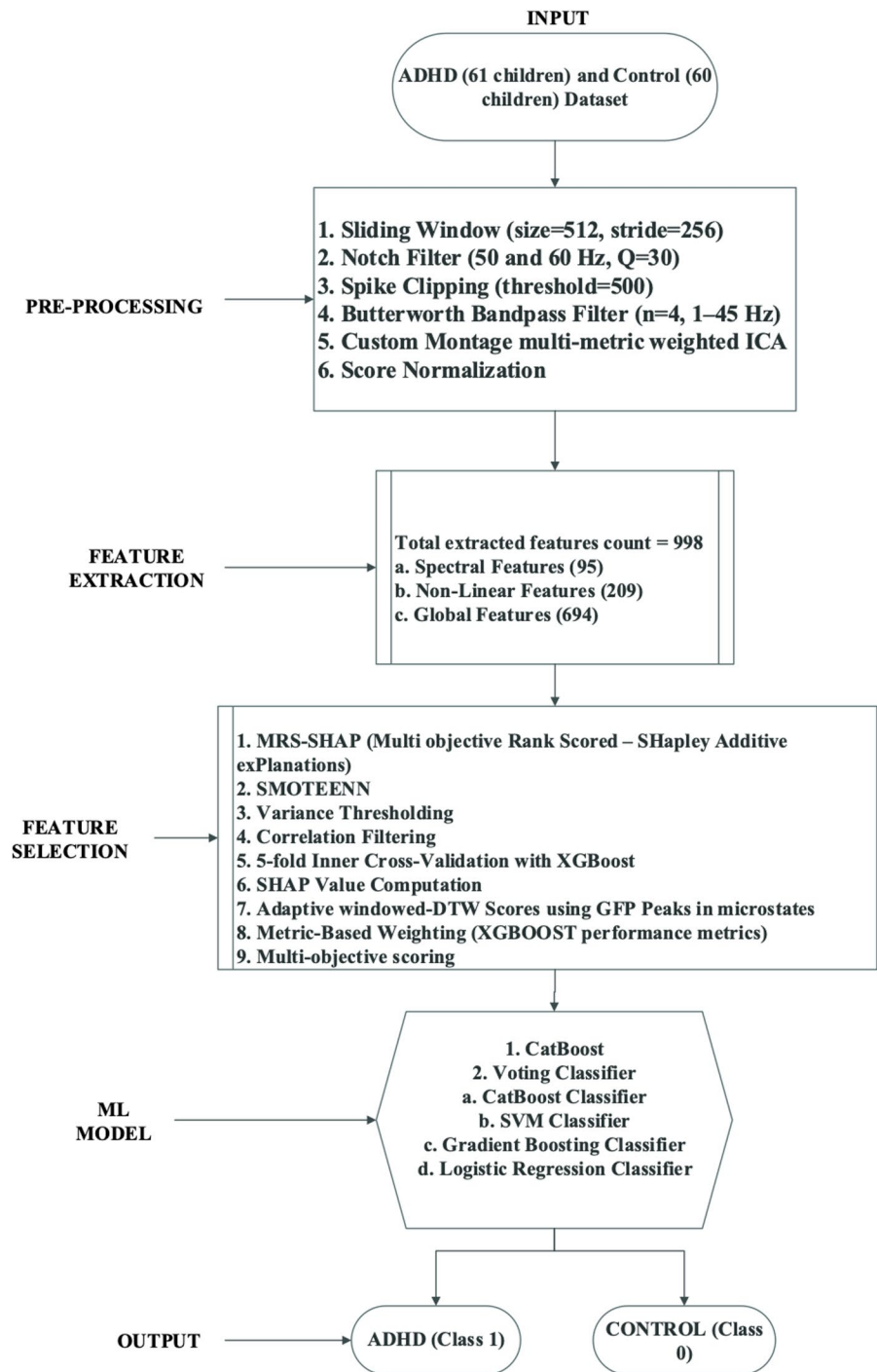


Fig. 1 A comprehensive workflow for ADHD vs. Control classification using EEG Data

MRS-SHAP (Multi-Objective Rank-Scored SHapley Additive exPlanations), integrates model explainability, temporal stability, and cross-validated robustness into a unified feature-selection strategy. This multi-objective integration ensures that the selected EEG features are not only statistically relevant but also physiologically explainable and temporally consistent, thereby improving both predictive accuracy and transparency. Two evaluation strategies were employed: (i) segment-level stratified cross-validation, and (ii) subject-level cross-validation using GroupKFold. The latter ensures that EEG segments

from the same subject do not appear in both training and testing sets, thereby preventing data leakage and providing a more realistic estimate of model generalization.

2.1 Dataset

The EEG dataset used in this study was obtained from the open-access repository “*EEG Data for ADHD / Control Children*” hosted on IEEE DataPort [24]. The dataset comprises 121 school-aged participants, including 61 children diagnosed with Attention Deficit Hyperactivity Disorder (ADHD) and 60 healthy controls. The ADHD group consisted of 48 boys and 13 girls (mean age: 9.62 ± 1.75 years), while the control group included 50 boys and 10 girls (mean age: 9.85 ± 1.77 years). All participants were between 7 and 12 years of age, corresponding to a critical developmental period for ADHD diagnosis. ADHD diagnoses were established by an experienced child and adolescent psychiatrist according to DSM-IV criteria at Roozbeh Hospital, Tehran, Iran. The dataset does not provide subtype-specific annotations (e.g., ADHD-IA, ADHD-HI, ADHD-C); therefore, ADHD is treated as a unified category in this study. Healthy control participants were screened to ensure no history of psychiatric disorders, epilepsy, or high-risk behaviors. Exclusion criteria for both groups included neurological disorders, brain injury (including epilepsy), major medical illness, learning or verbal disabilities, other psychiatric conditions, and the use of benzodiazepines or barbiturates. All participants were right-handed.

EEG recordings were acquired using a 19-channel system based on the international 10–20 electrode placement standard (Fz, Cz, Pz, C3, T3, C4, T4, Fp1, Fp2, F3, F4, F7, F8, P3, P4, T5, T6, O1, O2) at a sampling frequency of 128 Hz, with A1 and A2 electrodes serving as references. The data acquisition protocol was based on a task-driven visual attention paradigm. Participants were presented with a sequence of cartoon images containing 5–16 characters and were instructed to count the number of characters in each image. Each subsequent image was displayed immediately after the participant’s response, ensuring continuous engagement throughout the recording session. As a result, EEG recordings were not continuous full-day measurements but were obtained during controlled experimental sessions. The duration of each recording was variable and dependent on the participant’s response time, leading to recordings of differing lengths across subjects [24, 25]. It is noteworthy that the dataset exhibits a male-dominant distribution, particularly within the ADHD group, which aligns with epidemiological evidence indicating higher ADHD prevalence in boys during childhood. However, this imbalance may introduce bias in learned representations and should be considered when interpreting model generalizability.

2.2 Preprocessing

Preprocessing of EEG signals is essential for attenuating noise, artifacts, and interference, thereby improving the signal-to-noise ratio (SNR) and enhancing the performance of downstream machine-learning classifiers [26, 27]. In this work, a comprehensive preprocessing pipeline was developed to standardize recordings and preserve diagnostically relevant neural oscillations. First, all signals were standardized to a 19-channel configuration in accordance with the international 10–20 system [28]. Extreme amplitude values were normalized to maintain consistency across participants, considering the physiological EEG range of 10–100 μ V. To remove power-line interference, an IIR notch filter with

a Q-factor of 30 was primarily applied at 50 Hz, corresponding to the local power-line frequency of the recording environment. An additional 60 Hz notch filter was included as a precautionary measure to suppress any potential residual or equipment-induced interference. Power spectral inspection confirmed that the dominant noise component was centered at 50 Hz, and the inclusion of the 60 Hz notch did not introduce observable distortion in adjacent frequency bands. This approach ensured effective removal of line noise while preserving physiologically relevant EEG oscillations, particularly within the theta and beta bands [29, 30]. To illustrate the effectiveness of the preprocessing pipeline, representative EEG signals are shown at successive stages, including raw acquisition, notch filtering, and bandpass filtering. The figure demonstrates progressive attenuation of power-line interference, baseline drift, and high-amplitude artifacts while preserving physiologically relevant neural oscillations as shown Fig. 2.

Transient, high-amplitude artifacts such as those caused by muscle contractions, electrode pops, or amplifier saturation were mitigated through a two-stage spike-clipping procedure. A global threshold of $\pm 500 \mu\text{V}$ was used to suppress extreme transients without distorting genuine neural events, while a stricter $\pm 30 \mu\text{V}$ threshold was applied post-normalization to the Fp2 channel, which is particularly sensitive to ocular and frontal muscular activity [31, 32]. This channel-specific refinement was informed by the amplitude behavior of the adjacent Fp1 channel and was especially effective for ADHD recordings, which are prone to frequent motion-related artifacts. Subsequently, a fourth-order Butterworth band-pass filter (1–45 Hz) was applied to isolate physiologically relevant

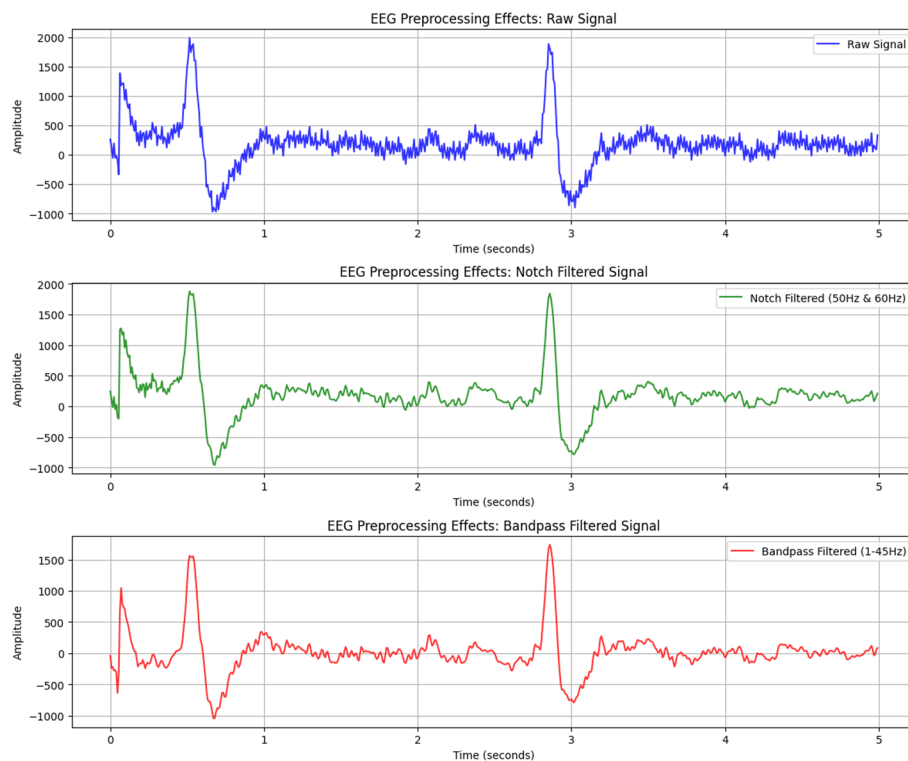


Fig. 2 Stepwise visualization of EEG preprocessing pipeline. Raw EEG signal (top), notch-filtered signal (middle), and bandpass-filtered signal (bottom) are shown for a representative channel segment. The notch filter (50 Hz and 60 Hz) effectively removes power-line interference, while the bandpass filter (1–45 Hz) suppresses low-frequency drift and high-frequency noise. The resulting signal exhibits improved stability and enhanced signal-to-noise ratio while preserving underlying neural activity

EEG bands delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz) while suppressing baseline drifts and high-frequency noise [33, 34]. The spatial configuration of the recording electrodes is shown in Fig. 3, illustrating comprehensive cortical coverage critical for ADHD-related neurophysiological pattern analysis.

To further suppress residual and complex artifacts, a multi-metric weighted Independent Component Analysis (ICA) was implemented, adapted from Nasrabadi et al. [24]. EEG signals were decomposed into 19 independent components, each evaluated through a composite artifact-scoring function that integrated four normalized metrics: kurtosis (0.5), variance (0.3), peak amplitude (0.15), and an Fp2-specific amplitude weight (0.05).

The weighting coefficients were determined based on established statistical characteristics of EEG artifacts and further validated through iterative inspection of artifact-prone EEG segments and guided by the relative sensitivity of each metric to distinct artifact types:

1. Kurtosis (0.5) was given the highest weight because it is highly sensitive to non-Gaussian, sharp transients such as eye blinks or muscle spikes dominant in pediatric ADHD EEG data [35–37].
2. Variance (0.3) captured sustained rhythmic disturbances and slow drifts arising from motion or electrode instability. Although informative, its contribution was moderated to avoid suppressing genuine oscillatory activity.
3. Peak amplitude (0.15) identified abrupt local deviations linked to amplifier saturation and electrode pops. Its smaller weight ensured that brief high-amplitude artifacts did not dominate the component ranking.
4. Fp2 amplitude (0.05) provided channel-specific sensitivity to ocular and frontal muscle artifacts while minimizing overcorrection in the prefrontal region, a site critical for ADHD-related cognitive control [31, 32].

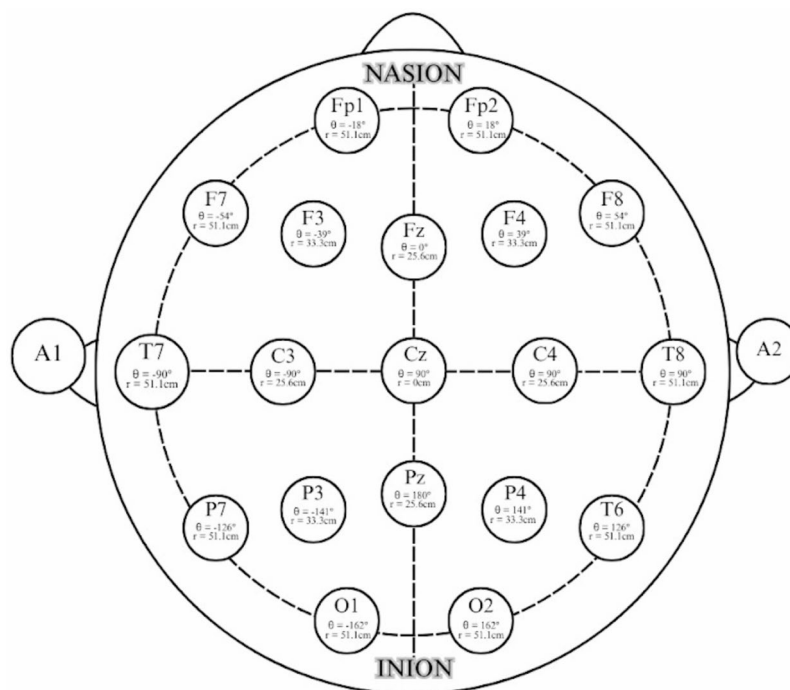


Fig. 3 Schematic representation of the international 10–20 EEG electrode placement system used in this study. The angular (θ) and radial (r) coordinates denote the spatial positioning of each electrode on the scalp

This composite weighting scheme was designed to balance artifact sensitivity with neural signal preservation. The dominance of kurtosis and variance aligns with prior ICA optimization studies that emphasize non-Gaussianity and signal dispersion as primary indicators of artifactual components [36, 38]. Peak amplitude contributes to detecting abrupt transient disturbances, while the inclusion of the Fp2-specific term accounts for the elevated susceptibility of frontal electrodes to ocular and muscle artifacts, particularly relevant in pediatric ADHD EEG recordings and weighting configuration was informed by established statistical properties of EEG artifacts and further validated through preliminary experiments, where it demonstrated improved artifact separation compared to unweighted and single-metric ICA approaches. Specifically, the proposed scheme achieved more consistent suppression of high-amplitude transients and frontal artifacts while preserving neurophysiologically relevant oscillatory activity within the theta and beta bands, which are critical for ADHD related analysis [30] and effectiveness of the weighted ICA approach is illustrated in Fig. 4, which compares raw and cleaned EEG signals across representative electrodes. The results show substantial artifact attenuation, particularly in frontal regions (Fp1, Fp2), while maintaining the spectral integrity of underlying neural signals required for reliable feature extraction and classification [39].

To evaluate neural signal characteristics post-cleaning and to explore ADHD-specific EEG features, multiple quantitative and spatial analyses were performed. Spectral entropy (SE), an index of signal irregularity, was computed for frontal regions (Fp1), revealing distinct distributional differences between ADHD and control groups as shown in the boxplot shown in Fig. 5 which exhibits a higher median value for ADHD patients compared to controlled subjects reinforcing the characteristic nature of increased randomness in ADHD EEG signal as compared to normal EEG signal. Topographic maps of beta power further delineated spatial variability, with ADHD subjects exhibiting heightened beta activity in frontal-central areas in Fig. 6, consistent with prior findings implicating hyperactivation and attention modulation disparities.

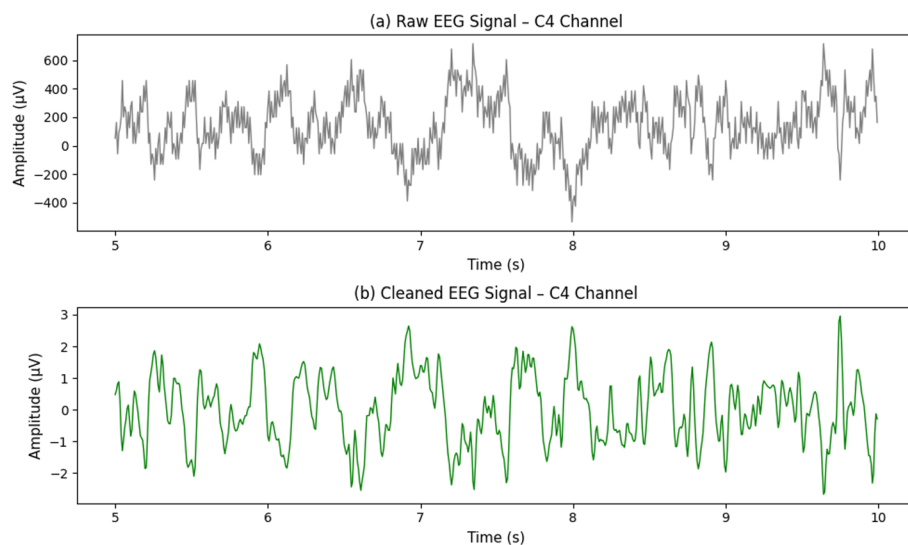


Fig. 4 Effectiveness of the proposed weighted ICA artifact removal approach EEG signals from the C4 channel over a 5–10 s segment are shown before (a) and after (b) preprocessing. The cleaned signal demonstrates substantial attenuation of high-amplitude artifacts and noise while preserving physiologically relevant neural activity

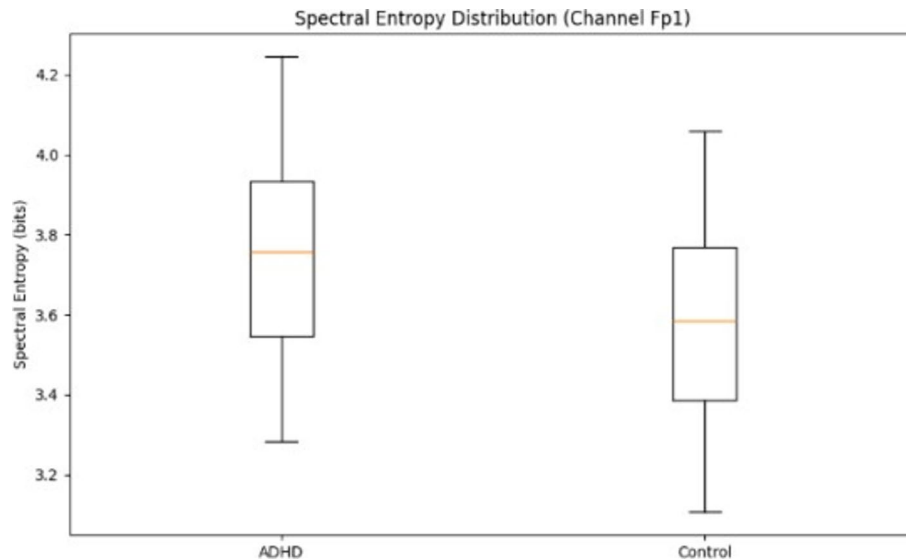


Fig. 5 Boxplot showing spectral entropy distribution at Fp1 for ADHD and control groups, indicating higher entropy variability in ADHD recordings

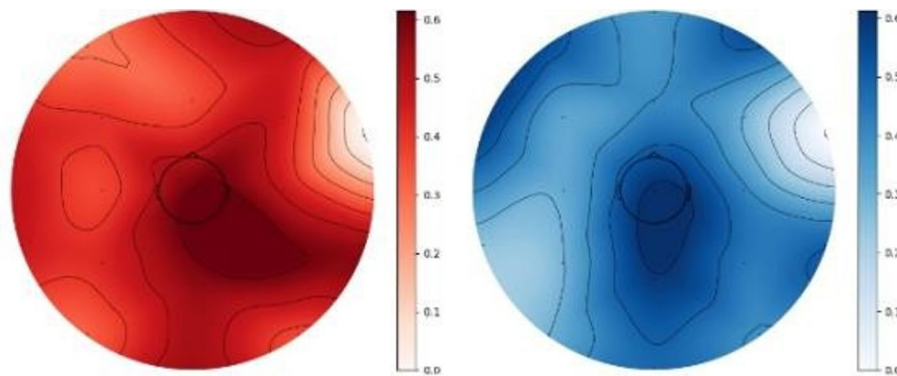


Fig. 6 Topographic beta power distribution in ADHD (left) and control (right) groups. Elevated frontal beta activity is evident in the ADHD cohort

The cross-spectral density (CSD) analysis was employed to examine functional coupling between bilateral frontal channels (Fp1-Fp2) within the theta band (4–8 Hz), uncovering a reduction in coherence for ADHD recordings relative to controls, as shown in Fig. 7. Furthermore, the normalized theta-beta ratio (TBR), a commonly investigated biomarker for ADHD, displayed broader interquartile variability in affected subjects in Fig. 8, suggesting altered spectral balance and developmental heterogeneity.

Elevated beta-band activity observed in the ADHD group aligns with prior findings; however, this pattern may also be influenced by stimulant medication effects, and therefore should be interpreted with caution.

2.3 Feature extraction

The feature extraction stage is critical in transforming the pre-processed EEG signals into a high-dimensional representation that captures the neurophysiological representations of brain activity. In this study, a total of 998 features were extracted and categorized

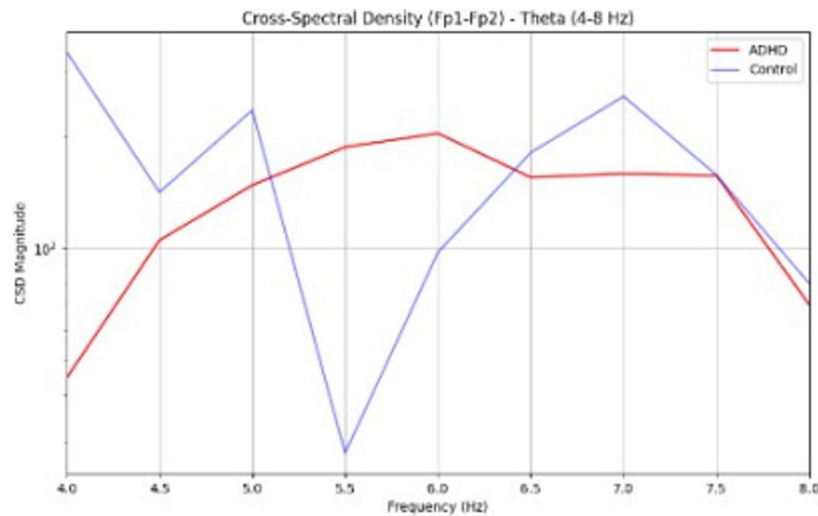


Fig. 7 Cross-spectral density plot of the Fp1- Fp2 theta band (4–8 Hz), illustrating reduced inter-hemispheric coherence in ADHD compared to control

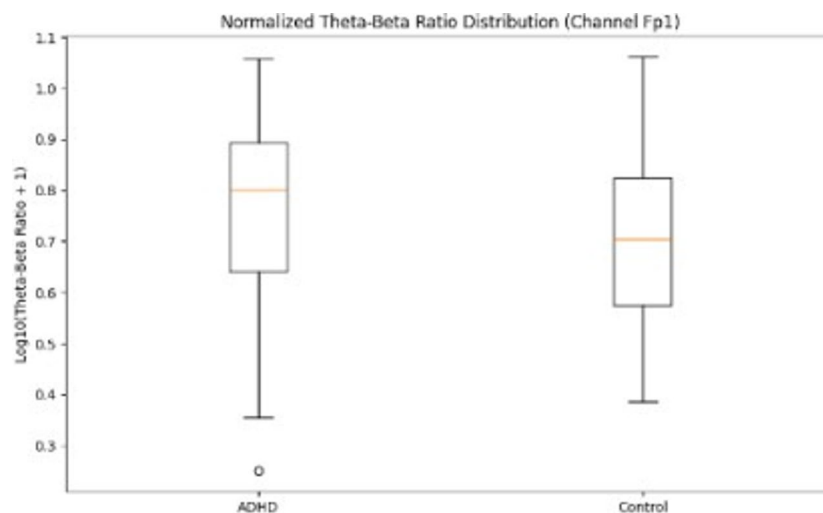


Fig. 8 Boxplot of normalized theta-beta ratio ($\log_{10}(\text{TBR} + 1)$) across ADHD and control groups. ADHD recordings show increased dispersion, aligning with neurodevelopmental instability hypotheses

into three main domains: Spectral (95 features), Nonlinear (209 features), and Global (694 features), as depicted in Fig. 9.

The extracted EEG features encompass spectral, temporal, and connectivity-based measures designed to capture neurophysiological patterns underlying attention deficit hyperactivity disorder (ADHD). Each feature provides a unique perspective on cortical activation, attentional control, and network-level synchronization. The selected features are summarized in Table 1 and elaborated below. The Engagement Index (EI) quantifies cognitive and emotional engagement by comparing beta activity (13–30 Hz) to the combined power of alpha (8–13 Hz) and theta (4–8 Hz) rhythms. High EI values indicate heightened cortical arousal, attentional focus, and information processing efficiency, while low EI reflects under-engagement or drowsiness [42, 43]. In ADHD, excessive theta and alpha activity paired with diminished beta oscillations produce a lower EI, reflecting prefrontal hypoactivation and inefficient top-down control [59]. These

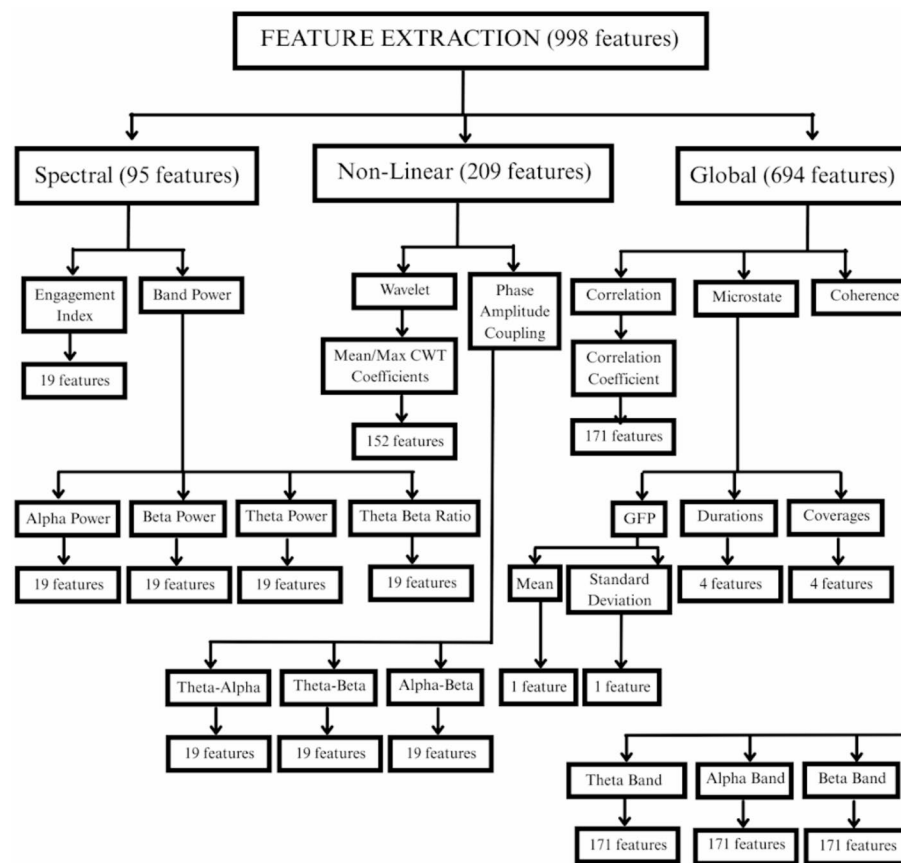


Fig. 9 Hierarchical breakdown of the 998 features extracted from EEG signals. Features are grouped into three primary categories: Spectral (95 features), Nonlinear (209 features), and Global (694 features), each subdivided based on frequency bands, time-frequency transformations, and connectivity metrics

abnormalities correlate with attention lapses, impulsivity, and reduced cognitive flexibility. Thus, EI serves as a simple yet physiologically explainable biomarker for attentional modulation and executive dysfunction in ADHD.

The Power Spectral Density (PSD) characterizes how signal energy is distributed across EEG frequencies, revealing dominant neural oscillations [41]. Using Welch's method, each EEG segment was windowed, transformed into the frequency domain, and averaged to reduce variance. PSD patterns in ADHD often exhibit increased theta (4–8 Hz) and reduced beta (13–30 Hz) power, especially in frontal regions associated with attention and inhibitory control [31]. This spectral imbalance signifies slower cortical rhythms and diminished readiness for task execution. By quantifying energy in specific bands, PSD enables direct correlation between oscillatory activity and neurobehavioral performance, serving as a foundation for higher-level indices such as EI and TBR is a classical biomarker for ADHD that reflects the relative strength of slow-wave (theta: 4–8 Hz) to fast-wave (beta: 13–30 Hz) activity [40].

Elevated TBR indicates cortical under-arousal and impaired vigilance. Numerous studies have reported significantly higher TBRs in ADHD subjects particularly in the frontal and central electrodes compared to healthy controls [44, 45]. This pattern mirrors neurodevelopmental delays in cortical maturation and deficits in executive functioning. Functionally, elevated TBR aligns with poor sustained attention, working-memory inefficiency, and hypoactivity within the prefrontal-striatal circuit. Consequently, TBR

Table 1 Spectral and Cross-Frequency EEG features: formulas and interpretations

Domain	Feature	Mathematical Expression	Description	Ref
Spectral	Power Spectral Density (PSD)	$\text{PSD } f = \frac{1}{N} \sum_{i=1}^N X_i(f) ^2 \quad (1)$	Frequency-domain representation of signal power distribution	[40]
	Band Power	$P_{band} = \frac{1}{F_2 - F_1} \int_{F_1}^{F_2} \text{PSD } f \, df \quad (2)$	Average power within EEG bands (delta, theta, alpha, beta)	[41]
	Engagement Index	$EI = \frac{P_{\theta} + P_{\alpha}}{P_{\beta}} \quad (3)$	Measures cognitive engagement and attentional processing	[42]
	Theta-Beta Ratio (TBR)	$TBR = \frac{\text{Average Theta Power}}{\text{Average Beta Power}} \quad (4)$	Marker indicating ADHD-related neural imbalance; high TBR correlates with attentional deficits	[43]
				[40]
Nonlinear	Phase-Amplitude Coupling (PAC)	$PAC = \left \frac{1}{N} \sum A(t) e^{j\varphi(t)} \right \quad (5)$	Cross-frequency coupling between phase and amplitude	[44]
	Mexican Hat Wavelet	$\phi(t) = (1 - t^2) e^{-t^2/2} \quad (6)$	Wavelet transform for detecting transient oscillations characteristic of ADHD brain activity	[45]
Global	Microstate Duration	$\text{Duration} = \frac{\text{Total time in state}}{\text{Number of occurrences}} \quad (7)$	Average persistence of EEG microstates	[46]
				[47]
	Channel Correlation	$CC_{ij} = \frac{\text{Cov}(x_i, x_j)}{\sigma_{x_i} \sigma_{x_j}} \quad (8)$ Where $\text{Cov}(x_i, x_j)$ is covariance between channels i and j, and $\sigma_{x_i}, \sigma_{x_j}$ are standard deviations.	Measures statistical synchrony between EEG channels, indicating network connectivity alterations in ADHD	[48]
Channel Coherence		$\text{Coh}_{ij}(f) = \frac{ S_{ij}(f) ^2}{S_{ii}(f)S_{jj}(f)} \quad (9)$ Where $S_{ij}(f)$ is the cross-spectral density, and $S_{ii}(f), S_{jj}(f)$ are auto-spectral densities.		[49]
				[50]
				[51]
Frequency-specific connectivity				[52]
				[53]
				[54]
ADHD-related metrics				[55]
				[56]
				[57]
Connectivity analysis				[58]
				[59]
				[60]

Note: α =alpha, β =beta, θ =theta, μ =mu, σ =sigma, ϕ =phi

remains one of the most reproducible and explainable electrophysiological markers of ADHD and Phase-Amplitude Coupling (PAC) captures nonlinear interactions between oscillations of different frequency bands, such as how the phase of a slow rhythm modulates the amplitude of a faster one [46]. Efficient PAC reflects coordinated communication between large-scale networks supporting perception, attention, and executive control. In ADHD, reduced theta-beta and alpha-beta PAC has been documented, indicating weakened coupling between cognitive control regions and sensorimotor pathways [47]. Such decoupling implies disrupted cross-frequency synchronization, impairing information integration across temporal scales. Hence, PAC serves as an advanced biomarker for quantifying temporal coordination deficits that underlie ADHD symptomatology.

The Mexican Hat Wavelet, defined as the second derivative of a Gaussian function, offers fine-grained temporal-frequency localization for analyzing non-stationary EEG signals [48]. Its symmetric shape and compact support make it particularly effective for detecting transient oscillations, bursts, and irregular rhythms common in ADHD. By decomposing EEG signals into localized time-frequency atoms, this wavelet reveals momentary shifts in theta-beta dominance, short-duration alpha suppression, and brief attention-related beta bursts [49]. These transient features provide critical insight into moment-to-moment neural instability a hallmark of ADHD pathophysiology that traditional Fourier-based metrics may overlook and EEG microstates represent brief, quasi-stable topographic patterns reflecting coordinated activity of large-scale neural networks [50]. Four canonical microstates (A-D) are typically observed, corresponding to auditory, visual, default-mode (DMN), and dorsal attention networks, respectively [51, 60]. In ADHD, microstate C (DMN) shows shortened duration and reduced coverage, consistent with abnormal self-referential processing and hypervariable resting-state dynamics [52]. Microstate D (Dorsal Attention Network) often appears more frequently but less stably, indicating excessive yet inefficient attentional engagement. The observed reduction in intra-state stability and increase in transition rate in ADHD reflect heightened neural variability and impaired network persistence neurophysiological correlates of distractibility and emotional dysregulation [53]. These findings reinforce the potential of microstates as non-invasive markers of ADHD-related network dysregulation.

Channel correlation quantifies linear dependencies between EEG signals recorded from distinct electrodes, serving as a proxy for large-scale functional connectivity. It is computed via the Pearson correlation coefficient across time samples, capturing synchronous amplitude fluctuations between cortical sites. In ADHD, reduced correlation particularly in the alpha band between frontal and parietal regions indicates weakened inter-regional communication and poor coordination within attentional and working-memory circuits [54, 55]. Lower connectivity within these networks reflects impaired top-down regulation and inconsistent cognitive control. By mapping correlation matrices, this feature provides insight into the network-level desynchronization that characterizes ADHD brain activity and Channel coherence extends the notion of correlation into the frequency domain, measuring the degree of phase-locked synchronization between signals across specific frequency bands [56]. Coherence values range from 0 (no synchrony) to 1 (complete synchronization). ADHD studies consistently report abnormal coherence patterns notably increased short-range frontal coherence (linked to impulsivity) and decreased fronto-parietal coherence (indicative of poor attentional

regulation) [57, 58]. Such spectral desynchronization reflects disrupted oscillatory communication between executive and sensory networks. Therefore, coherence analysis complements correlation metrics by elucidating frequency-specific connectivity deficits underlying ADHD. Together, these EEG-derived features provide a multidimensional representation of ADHD-related neural activity, integrating spectral, temporal, and connectivity domains to support explainable machine learning-based classification and interpretation.

The selection of EEG channels and frequency bands in this study is not arbitrary but is implicitly driven by the feature selection framework. Rather than predefining specific channels or bands, the proposed MRS-SHAP approach evaluates all extracted features across channels and frequency sub-bands and ranks them based on their contribution to classification performance. SHAP-based importance analysis revealed that features derived from specific channels and frequency bands consistently exhibited higher discriminative power. In particular, frontal and central regions, along with beta and theta frequency bands, contributed most significantly to ADHD classification. This aligns with established neurophysiological findings linking frontal dysfunction and altered theta–beta dynamics with attentional deficits. Thus, channel and band selection in this study emerges from a data-driven, performance-based ranking process rather than manual or heuristic assumptions, ensuring objective identification of the most informative EEG features.

2.4 Feature selection

Feature selection plays a pivotal role in EEG-based machine learning applications by identifying the most informative subset of features from a high-dimensional feature space. EEG signals are inherently complex and multifaceted, often producing a large number of extracted features that may include redundancy and noise. The inclusion of such features increases computational complexity, heightens the risk of overfitting, and reduces interpretability and generalizability. Traditional feature-selection approaches can be broadly categorized into wrapper and filter methods. Wrapper methods, such as Recursive Feature Elimination (RFE), forward selection, and backward elimination, iteratively evaluate subsets of features based on model performance. Filter methods, including ANOVA, Pearson's correlation, chi-square test, and Mutual Information, assess individual features according to their statistical relevance to the target variable. Although effective in certain contexts, these approaches often rely on a single selection criterion either statistical relevance or predictive strength and may overlook crucial feature interactions, which are particularly significant in EEG data due to their spatiotemporal dependencies.

To address these limitations, a novel feature-selection algorithm termed Multi-objective Rank-Scored SHapley Additive exPlanations (MRS-SHAP) was employed. This approach integrates multiple evaluation criteria within a unified scoring framework that simultaneously accounts for feature relevance, interaction strength, and temporal dynamics. By incorporating explainability through SHAP values [61], MRS-SHAP enables the identification of features that are not only highly predictive but also biologically explainable within the context of ADHD-related neurophysiology. From an initial pool of 998 extracted EEG features spanning spectral, temporal, statistical, and

connectivity domains, the MRS-SHAP algorithm derives an optimal subset that maximizes classification performance while retaining neurobiological interpretability.

Given the inherent class imbalance between ADHD and control samples, the Synthetic Minority Oversampling Technique combined with Edited Nearest Neighbors (SMOTE-ENN) was applied. SMOTE-ENN improves upon standard SMOTE by both generating synthetic minority-class samples and cleaning ambiguous or noisy data. Initially, SMOTE interpolates between existing minority instances to balance class proportions. However, synthetic oversampling can introduce overlapping or outlier points that degrade model generalization. To mitigate this, the ENN component removes samples whose class labels differ from the majority of their k -nearest neighbors. This hybrid approach enhances the representativeness of minority samples, minimizes class overlap, and yields a cleaner training dataset ultimately improving classifier accuracy and generalizability in ADHD classification tasks. Following data balancing, variance and correlation thresholding were applied for redundancy reduction. A variance threshold of 0.01 was selected to remove low-variability (near-constant) features, which contribute minimal discriminative information and can negatively impact model stability. Next, a Correlation Threshold of 0.85 was applied to remove highly correlated features. The Pearson correlation coefficient between any two features i and j was computed as:

$$R_{ij} = \frac{\text{cov}(f_i, f_j)}{\sigma_{f_i} \sigma_{f_j}} \quad (10)$$

where $\text{cov}(f_i, f_j)$ is the covariance, and σ_{f_i} and σ_{f_j} are the standard deviations of features f_i and f_j , respectively.

For any feature pair exceeding this correlation threshold, one feature was removed. This dual-step pruning retained only non-redundant, information-rich features, improving both interpretability and efficiency of subsequent SHAP-based importance scoring. The filtered feature set was then evaluated using a SHapley Additive exPlanations (SHAP)-based importance ranking. While SHAP is conventionally used for post-hoc model interpretation, in this study it was adapted as a *feature-selection tool*, enabling the ranking and retention of features that contribute most to predictive accuracy and model transparency. To generate SHAP values, an initial eXtreme Gradient Boosting (XGBoost) model was trained using the pruned feature set. XGBoost, a scalable and regularized gradient-boosting algorithm, constructs an ensemble of sequential decision trees optimized via second-order gradient information. It incorporates both L1 and L2 regularization to prevent overfitting and performs well with high-dimensional EEG data. Its proven ability to capture nonlinear dependencies in EEG features was demonstrated in [61], achieving ~99% accuracy in multi-class emotion classification. For the present work, a five-fold cross-validated XGBoost model was implemented. The hyperparameters used for model tuning and preliminary prediction are summarized in Table 2. To prevent data leakage, all preprocessing and feature selection steps were performed strictly within the training data of each cross-validation fold. Specifically, SMOTE-ENN resampling, SHAP-based feature importance computation, and MRS ranking were applied exclusively on the training fold. The test fold remained completely unseen during feature selection and model training. This nested cross-validation strategy ensures

Table 2 Optimized XGBoost hyperparameters for EEG-based ADHD classification balancing complexity and performance

HYPERPARAMETERS	VALUES
n_estimators	150
max_depth	6
min_child_weight	2
gamma	0.5
reg_alpha, reg_lambda	0.5,1.0
eval_metric	logloss
scale_pos_weight	class_weights[1]/class_weights[0]

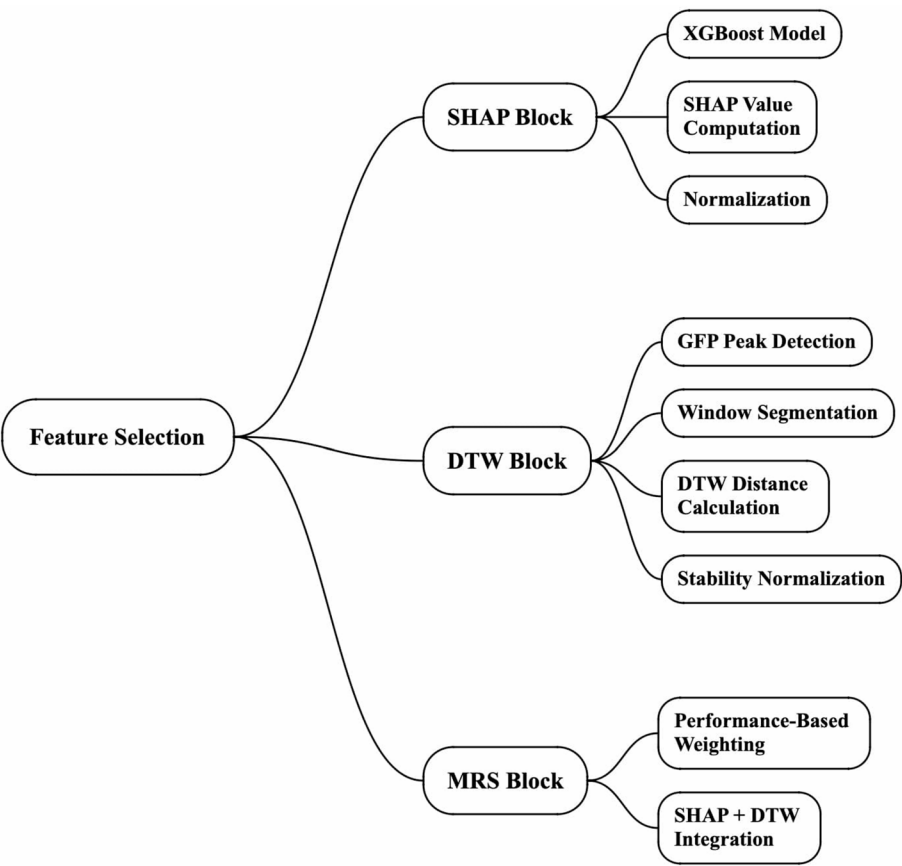


Fig. 10 Workflow of SHAP-based importance estimation within the proposed MRS-SHAP feature selection framework

unbiased performance estimation and prevents information leakage between training and testing data.

The SHAP-based feature importance computation process is illustrated in Fig. 10 Initially, an XGBoost model is trained using the filtered feature set, after which SHAP values are computed using a TreeExplainer to quantify the contribution of each feature to the model predictions. The mean absolute SHAP values across all samples are then calculated to obtain global feature importance scores, which are subsequently normalized to ensure comparability across features. These normalized SHAP scores are later integrated into the proposed multi-objective ranking framework.

2.4.1 SHAP-based importance and multi-objective ranking

After model execution, SHAP values were computed using a TreeExplainer, which quantifies the contribution of each feature f_i to the model's prediction for each sample x_j ($j = 1, 2, \dots, m$, where m denotes the total number of samples). The SHAP value $\text{SHAP}_{i,j}$ represents the marginal change in model output when feature f_i is included compared to its exclusion from all possible feature subsets. Formally, it is defined as:

$$\text{SHAP}_{i,j} = \sum_{S \subseteq F_{\text{filtered}} \setminus \{f_i\}} \frac{|S|!(n-|S|-1)!}{n!} [g(x_{S \cup \{f_i\}}) - g(x_S)] \quad (11)$$

where S is a subset of the feature set, n is the total number of features, and $g(x_S)$ denotes the model's prediction when only the features in subset S are considered. The mean absolute SHAP value across all samples provides the global importance score of each feature f_i :

$$\text{SHAP}_i = \frac{1}{m} \sum_{j=1}^m |\text{SHAP}_{i,j}| \quad (12)$$

To ensure comparability among features, SHAP scores were **normalized** within the range $[0, 1]$ using:

$$\text{SHAP}_{\text{norm},i} = \frac{\text{SHAP}_i - \min(\text{SHAP})}{\max(\text{SHAP}) - \min(\text{SHAP}) + \epsilon} \quad (13)$$

where ϵ is a small constant for numerical stability. This normalization allows the SHAP-based importance scores to be seamlessly integrated into the multi-objective ranking framework.

2.4.2 Temporal stability via dynamic time warping (DTW)

To analyze temporal consistency and the stability of EEG-derived features, Dynamic Time Warping (DTW) was employed. DTW measures the optimal alignment cost between temporal windows derived from microstate-based EEG features, capturing temporal deformation patterns across samples. The Global Field Power (GFP) was used to estimate signal variance and identify characteristic peaks, and adaptive windowing was applied using the mean inter-peak interval of microstate peaks. A fallback window of 50 samples was used when peak detection was insufficient.

The average pairwise DTW distance across all sub-windows for a given feature was computed as:

$$\text{DTW}_i = \frac{1}{KC_i} \sum_{k=1}^K \sum_{l=k+1}^K \text{DTW}(W_{i,k}, W_{i,l}) \quad (14)$$

where K is the number of sub-windows, C_i represents the number of valid comparisons, and

$DTW(W_{i,k}, W_{i,l})$ denotes the minimum-cost alignment path based on Euclidean distances. The resulting DTW distances were inverted and normalized to yield stability scores, ensuring that temporally consistent features received higher values:

$$DTW_{\text{norm},i} = \frac{\max(DTW) - DTW_i}{\max(DTW) - \min(DTW) + \epsilon} \quad (15)$$

2.4.3 Integration into multi-objective ranking score (MRS)

To unify SHAP importance, temporal stability, and model performance, Multi-Objective Ranking Score (MRS) was formulated. Model performance feedback was incorporated via a balanced weight derived from inner-loop five-fold cross-validation metrics: Sensitivity, Specificity, F1-score, and Accuracy. Each metric m was normalized and inverted as:

$$\text{metric}_{\text{norm},m} = \frac{m - \min(m)}{\max(m) - \min(m) + \epsilon} \quad (16)$$

$$\text{weight}_m = 1 - \text{metric}_{\text{norm},m} \quad (17)$$

The balanced weight (BW) was then computed as the average of all four metric weights:

$$BW = \frac{1}{4} \sum_{m=1}^4 \text{weight}_m \quad (18)$$

Finally, the Multi-Objective Ranking Score (MRS) for each feature was calculated as a weighted combination of normalized SHAP and DTW scores, where suboptimal model performance amplifies SHAP weighting:

$$MRS_i = 0.6 \times \text{SHAP}_{\text{norm},i} \times (1 + BW) + 0.4 \times \text{DTW}_{\text{norm},i} \quad (19)$$

Based on the ranked MRS values, the top 60% of features were retained as the final subset for ADHD classification. This framework integrates variance, correlation, SHAP-derived importance, temporal consistency, and performance-based weighting into a cohesive multi-objective pipeline.

It ensures that the selected features are both highly discriminative and neurophysiologically explainable, thus enhancing model performance and clinical explainability. A comprehensive schematic of the proposed feature-selection pipeline is illustrated in Fig. 11.

3 Results and discussion

This section presents and discusses the outcomes of applying the proposed Multi-Objective Rank Score SHapley Additive exPlanations (MRS-SHAP) feature selection framework to the ADHD EEG dataset. The analysis evaluates classification performance, model robustness, and interpretability of the selected features in relation to established neurophysiological biomarkers of ADHD. The preprocessing pipeline integrated

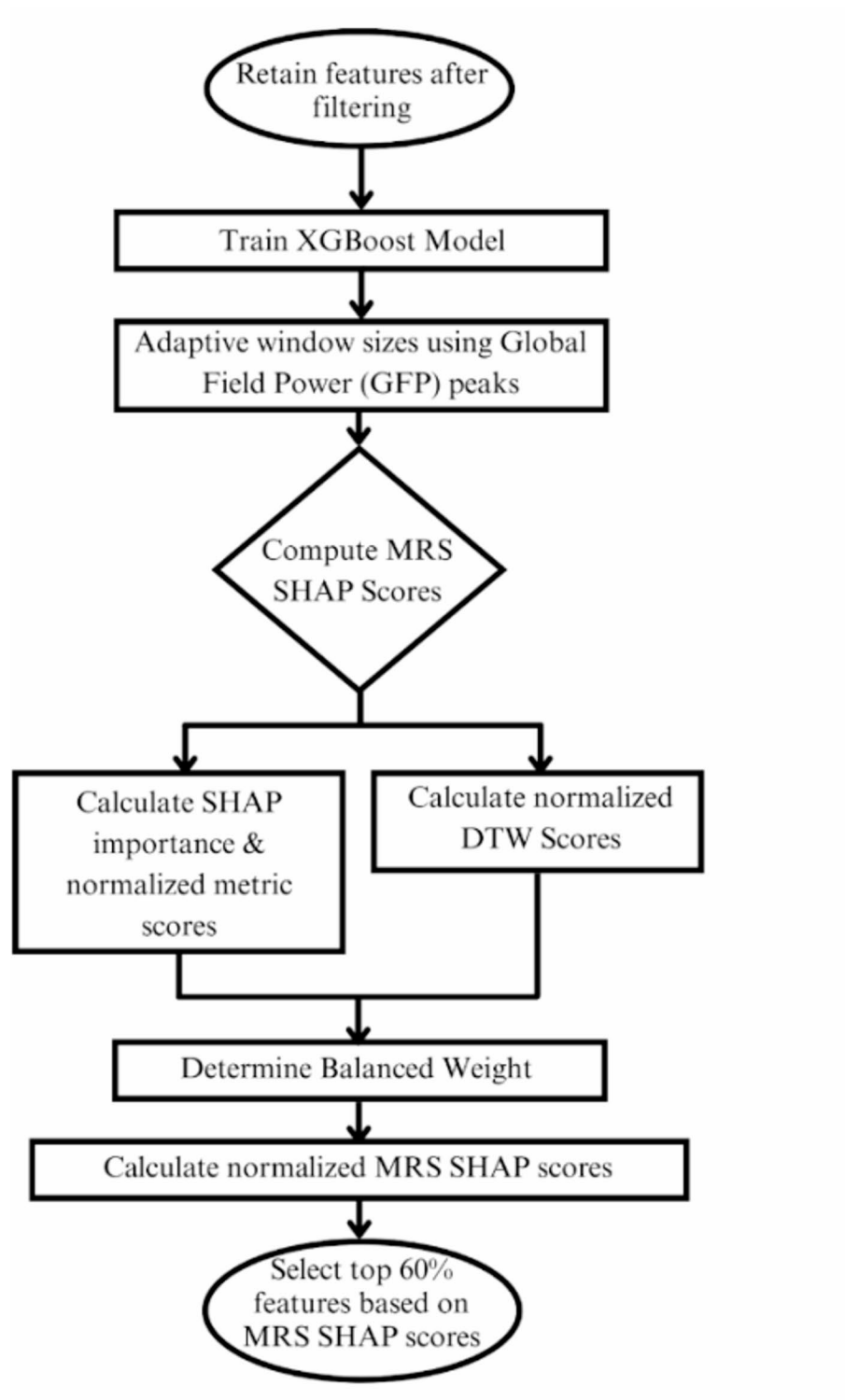


Fig. 11 Feature selection workflow integrating MRS SHAP scores to determine balanced-weight (BW) metrics for optimal EEG feature subset selection

multi-level filtering and adaptive weighting to suppress ocular, muscular, and environmental artifacts. Variance thresholding eliminated low-variance, non-informative features, while correlation-based filtering reduced redundancy from co-linear EEG channels. A custom multi-metric weighting scheme combining kurtosis (0.5), variance (0.3), peak amplitude (0.15), and Fp2 amplitude (0.05) further improved artifact suppression, ensuring enhanced signal fidelity and stability. This modular design supports transferability across different EEG datasets through weight re-optimization, providing a scalable foundation for standardized EEG preprocessing. Conventional EEG-based ADHD studies often employ static, unidimensional feature extraction methods such as simple spectral ratios or Principal Component Analysis (PCA) which inadequately capture ADHD's dynamic neural heterogeneity [50, 62, 63]. Such techniques typically produce either oversimplified or excessively large feature sets that mask transient but clinically meaningful neural fluctuations [64–66]. The proposed pipeline addresses this gap via adaptive microstate segmentation, leveraging Global Field Power (GFP) peaks to dynamically define windows that align with natural neural transitions. This adaptation enables the detection of short-lived but informative EEG dynamics that traditional fixed-epoch approaches often miss.

3.1 Multi-objective feature selection strategy

At the core of the MRS-SHAP pipeline lies a multi-objective optimization framework combining three criteria:

1. Feature diversity - Ensuring inclusion from spectral, temporal, and connectivity domains.
2. Temporal stability - Measured using Dynamic Time Warping (DTW)-based alignment.
3. Model importance - Captured through SHAP-derived feature relevance with multi-metric performance weighting.

This tri-criteria design prevents over-reliance on a single domain, yielding a compact yet physiologically rich feature subset robust to variations in EEG montage, acquisition hardware, and experimental context. Temporal instability a hallmark of ADHD characterized by heightened neural variability is explicitly addressed by incorporating DTW-based stability scoring.

By emphasizing temporal consistency, the selected features remain discriminative and generalizable across sessions and subjects, capturing ADHD's temporal dysregulation with superior fidelity compared to prior works [67, 68].

3.2 Comparative evaluation with baseline methods

The MRS-SHAP framework was benchmarked against established dimensionality reduction and feature selection techniques, including PCA, ANOVA, Forward Selection, Information Gain, and Recursive Feature Elimination (RFE). Classification was performed using both CatBoost and a soft Voting Ensemble (CatBoost-SVM-Gradient Boosting-Logistic Regression) with optimized weights (0.3, 0.3, 0.25, 0.15), as presented in Table 3. A qualitative comparison of the evaluated feature selection methods is further summarized in Table 4.

Table 3 Performance comparison of MRS-SHAP against baseline methods

Method	Model	Sensitivity	Specificity	F1-Score	Accuracy
Proposed - MRS-SHAP	CatBoost	0.9284	0.9345	0.9379	0.9310
	Voting	0.9135	0.9273	0.9358	0.9285
PCA	Voting	0.9072	0.9072	0.9018	0.8891
ANOVA	Voting	0.9072	0.8608	0.9000	0.8868
Forward Selection	Voting	0.9111	0.8472	0.8975	0.8831
Information Gain	Voting	0.9111	0.8947	0.9142	0.9039
RFE	Voting	0.9294	0.9072	0.9018	0.8891

MRS-SHAP outperformed all baselines improving accuracy by ~7% over PCA, ~5% over ANOVA, and ~4% over Forward Selection. These results highlight the discriminative strength and stability of the selected features

Table 4 Qualitative comparison of EEG-based feature selection methods

Method	Biological Relevance	Temporal Sensitivity	Feature Diversity	Interpretability	Computational Complexity
MRS-SHAP	High	High (DTW-based)	High	High (SHAP explainability)	Moderate
ANOVA	Low	Low	Low	Moderate	Low
PCA	Low	Low	Moderate	Low	Low
RFE	Moderate	Low	Moderate	Low	High
Forward Selection	Low	Low	Low	Moderate	High
Information Gain	Low	Low	Low	Moderate	Low

MRS-SHAP achieved superior temporal sensitivity, interpretability, and biological consistency, validating its multi-objective design, as shown in Table 4

Table 5 CatBoost hyperparameter search space

Parameter	Search Range / Values	Description
Iterations	[200, 300, 400]	Number of boosting trees
Depth	[4, 6, 8]	Tree depth controlling model complexity
Learning Rate	[0.01, 0.05, 0.1]	Step size for gradient boosting
L2 Leaf Regularization	[3, 5, 7, 10]	Regularization to prevent overfitting
Border Count	[64, 128, 254]	Number of splits for numerical features
Bagging Temperature	[0, 0.5, 1]	Controls randomness in sampling
Random Strength	[1, 2, 5]	Adds randomness to tree splits
Class Weights	Balanced / None	Handles class imbalance
Early Stopping Rounds	[30, 50]	Stops training when no improvement
Loss Function	Logloss	Binary classification objective
Eval Metric	AUC / F1	Performance evaluation metric

3.3 Qualitative comparison of feature selection methods

3.4 Model optimization and validation

Hyperparameter optimization for the CatBoost classifier was performed using RandomizedSearchCV over a predefined search space, as summarized in Table 5. The selected ranges were chosen to balance model complexity, generalization capability, and computational efficiency. Parameters such as tree depth, learning rate, and L2 regularization were tuned to mitigate overfitting, while stochastic parameters including bagging temperature and random strength improved robustness. Early stopping was employed to prevent unnecessary training iterations and ensure optimal convergence.

A stratified five-fold cross-validation was initially performed at the segment level as shown Table 6. The CatBoost classifier achieved an average accuracy of $98.40\% \pm 0.62\%$

Table 6 Five-fold cross-validation of CatBoost

Fold	Sensitivity	Specificity	F1-Score	Accuracy
1	0.9916	0.9821	0.9877	0.9877
2	0.9937	0.9881	0.9914	0.9914
3	0.9811	0.9792	0.9803	0.9803
4	0.9874	0.9851	0.9864	0.9864
5	0.9832	0.9613	0.9741	0.9741
Average Performance (Mean $\pm$ SD)	0.9874 $\pm$ 0.0048	0.9792 $\pm$ 0.0102	0.9840 $\pm$ 0.0062	0.9840 $\pm$ 0.0062

Table 7 Statistical Validation and Performance Significance

Metric	Model	Mean $\pm$ SD	95% CI
Accuracy	CatBoost	0.9840 $\pm$ 0.0062	[0.9763, 0.9917]
F1-score	CatBoost	0.9840 $\pm$ 0.0062	[0.9763, 0.9917]
Sensitivity	CatBoost	0.9874 $\pm$ 0.0048	[0.9814, 0.9934]
Specificity	CatBoost	0.9792 $\pm$ 0.0102	[0.9665, 0.9919]
Accuracy	Voting Ensemble	0.9285 $\pm$ 0.0052	[0.9221, 0.9350]
F1-score	Voting Ensemble	0.9358 $\pm$ 0.0053	[0.9293, 0.9424]

and an F1-score of $98.40\% \pm 0.62\%$, indicating strong discriminative performance. However, it is important to note that segment-level splitting allows EEG segments from the same subject to appear in both training and testing sets. This may introduce optimistic bias due to intra-subject data leakage, as the model can learn subject-specific patterns rather than generalizable neurophysiological features. Therefore, while these results demonstrate the upper-bound performance of the proposed framework, they should be interpreted with caution. To obtain a more realistic estimate of generalization, a strict subject-level evaluation was conducted, as described in the subsequent section.

Statistical significance testing across cross-validation folds was not performed, as folds are not independent samples. Instead, performance is reported using mean and standard deviation across folds to provide a robust descriptive evaluation.

3.5 Statistical validation and performance significance

To quantify the reliability of the reported classification performance, statistical summaries were computed from the five-fold cross-validation results for the CatBoost classifier. For each performance metric (Accuracy, F1-score, Sensitivity, and Specificity), the mean, standard deviation (SD), and 95% confidence intervals (CI) were estimated using the Student's *t*-distribution ($df=4$). It is important to note that statistical hypothesis testing across cross-validation folds was not performed, as folds are not independent samples. Performing paired *t*-tests on fold-wise results may lead to invalid inferences due to dependency between splits. Therefore, performance evaluation is reported using descriptive statistics (mean $\pm$ SD and confidence intervals), which provide a more appropriate and robust assessment of model stability.

The results indicate consistently high performance with low variance across folds, demonstrating the stability of the CatBoost classifier under segment-level evaluation as shown Table 7. However, as discussed in Sect. 3.4, these results may reflect an upper-bound estimate due to potential intra-subject similarity in segment-level splitting. To ensure a more realistic assessment of generalization, subject-level validation results are additionally reported in Sect. 3.6. These results provide a stricter evaluation setting and better reflect real-world deployment conditions. Furthermore, ablation analysis (Sect.

3.12) was conducted to evaluate the contribution of individual components of the proposed MRS-SHAP framework. The combined approach demonstrated improved robustness compared to individual feature selection strategies, supporting the effectiveness of the multi-objective design.

3.6 Subject-level cross-validation performance

To further assess model generalization and eliminate potential subject-level data leakage, a GroupKFold-based subject-level cross-validation was performed, ensuring that EEG segments from the same subject were not shared between training and testing sets. As shown in Table 8, the model achieved an average accuracy of 72.61% and F1-score of 76.36%, which is lower than segment-level evaluation but reflects a more realistic estimation of real-world performance. The observed performance drop highlights the impact of inter-subject variability in EEG signals and confirms that segment-level evaluation may overestimate model performance when subject overlap is not controlled. This analysis strengthens the robustness of the proposed framework by demonstrating its behavior under strict subject-wise validation.

This subject-level validation directly addresses potential data leakage concerns and provides a more clinically relevant estimate of model generalization.

3.7 Model explainability through SHAP analysis

To interpret the model’s decision-making process and identify the neurophysiological basis of its predictions, SHapley Additive exPlanations (SHAP) analysis was conducted. Figure 12 presents the SHAP summary plot for the CatBoost classifier, highlighting the most influential features across all EEG samples and analysis revealed that features derived from frontal and central regions, particularly within the theta and beta frequency bands, contributed most strongly to model predictions consistent with neural circuits involved in attention and executive control, which are known to be disrupted in ADHD. Among the top-ranked features, beta_ch18 and corr_ch4_ch18 exhibited the highest mean SHAP values (≈ 0.5), followed by corr_ch0_ch2 (≈ 0.4), corr_ch2_ch10 and corr_ch10_ch11 ($\approx 0.35 - 0.3$), and beta_ch13 and corr_ch0_ch10 (≈ 0.25). Additional features such as wavelet_mean1_ch4, corr_ch7_ch9, and corr_ch3_ch7 showed moderate importance ($\approx 0.2 - 0.15$), while low-impact features clustered near zero, indicating minimal contribution. Physiologically, connectivity features such as corr_ch4_ch18 suggest altered occipital–central interactions, while beta-band activity in occipital regions (e.g., O1, O2) reflects mechanisms related to sustained attention and visual processing both commonly impaired in ADHD. The prominence of both connectivity-based and spectral features highlights the importance of integrating multi-domain EEG characteristics.

To provide quantitative evidence for channel and frequency band relevance, SHAP values were further aggregated across electrode locations and frequency bands (delta,

Table 8 Subject-Level Cross-Validation Performance (GroupKFold, 5-fold)

Fold	Sensitivity	Specificity	F1-Score	Accuracy
1	0.8046	0.8317	0.8455	0.8146
2	0.9471	0.6475	0.7925	0.7802
3	0.8551	0.2508	0.7300	0.6168
4	0.7835	0.8172	0.8116	0.7985
5	0.4822	0.9350	0.6384	0.6200
Mean ± SD	0.7745 ± 0.17	0.6964 ± 0.26	0.7636 ± 0.08	0.7261 ± 0.09

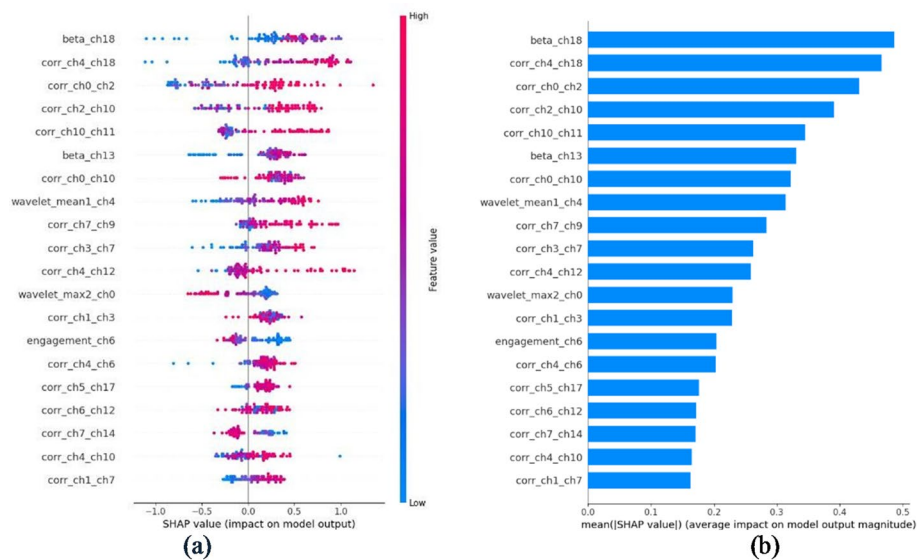


Fig. 12 SHAP summary plot for CatBoost model showing feature impact on output predictions **(a)** SHAP bee swarm plot showing magnitude and direction of feature effects and **(b)** SHAP bar plot showing the average absolute value for each feature influence

Table 9 Summary of machine learning models and key hyperparameters tested in the ensemble Voting Classifier for ADHD classification

Model	Key Parameters Tested
CatBoost	Iterations [200, 300], Depth [6, 8], Learning Rate [0.05, 0.1], L2 Leaf Reg [5, 10], Border Count [128]
SVM	C [10, 50], Kernel ['rbf'], Gamma ['scale', 0.01]
Gradient Boosting	n_estimators [200, 300], Max Depth [5, 7], Learning Rate [0.05, 1]
Logistic Regression	C [0.1, 1, 10], Penalty ['l2'], Solver ['liblinear']

theta, alpha, beta). Features were grouped accordingly, and their mean contributions were computed. This analysis demonstrated that frontal (Fp1, Fp2, F3, F4) and central (C3, C4) electrodes consistently exhibited higher importance scores compared to posterior regions. In terms of frequency bands, theta and beta bands showed the strongest discriminative power, whereas alpha and delta bands contributed less prominently. These findings confirm that the model's feature selection is data-driven rather than heuristic, and they align with existing neurophysiological evidence indicating that ADHD is characterized by altered frontal-central dynamics and theta-beta activity patterns.

3.8 Ensemble voting classifier evaluation and SHAP interpretation

The selected features were further evaluated using a five-fold cross-validated ensemble Voting Classifier comprising CatBoost, Support Vector Machine (SVM), Gradient Boosting (GB), and Logistic Regression (LR). A soft voting strategy was implemented with optimized fixed weights (0.3, 0.3, 0.25, and 0.15, respectively), allowing the ensemble to leverage the complementary strengths of the individual models. Hyperparameter optimization for each classifier was conducted via RandomizedSearchCV, ensuring robust generalization and enhanced predictive performance. The search ranges used for each model are summarized in Table 9.

This ensemble configuration demonstrated high predictive robustness, with consistent performance across all folds, validating the reliability of MRS-SHAP-selected features.

Table 10 Comparative performance metrics of Voting Classifier across five cross-validation folds

Fold	Sensitivity	Specificity	F1-Score	Accuracy
Fold 1	0.9125	0.9440	0.9329	0.9263
Fold 2	0.9284	0.9270	0.9352	0.9278
Fold 3	0.9257	0.9185	0.9307	0.9226
Fold 4	0.9589	0.9083	0.9445	0.9367
Fold 5	0.9216	0.9389	0.9359	0.9292

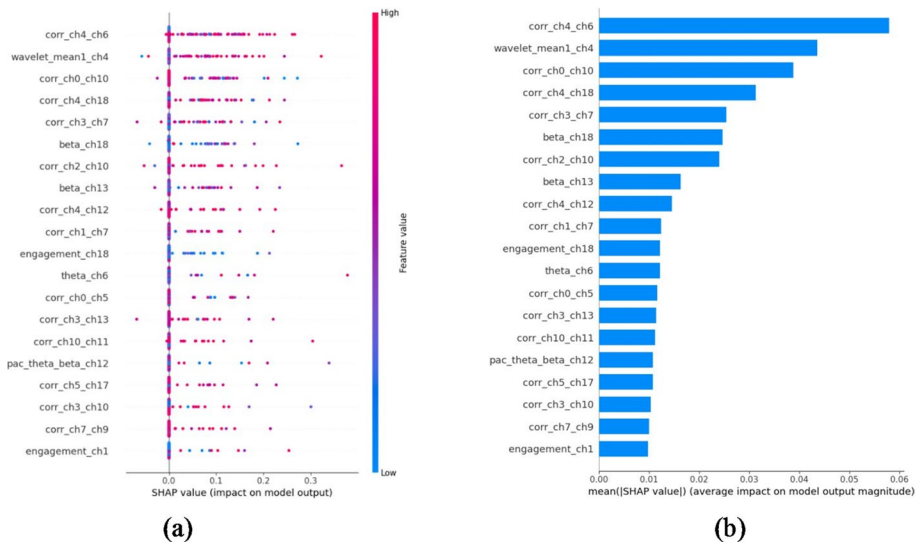


Fig. 13 SHAP Plot Summary for Voting Classifier showing feature impact on output predictions **(a)** SHAP bee swarm plot showing magnitude and direction of feature effects and **(b)** SHAP bar plot showing the average absolute value for each feature influence

The model achieved peak F1-scores of 94.45% and accuracy of 93.67%, indicating strong stability and discriminative ability across folds as shown Table 10.

To interpret the ensemble’s decision process, SHAP (SHapley Additive exPlanations) analysis was employed Fig. 13. The SHAP summary plot revealed consistent feature importance trends across base learners, confirming cross-model stability of the selected features. The most influential features included corr_ch4_ch6 (≈ 0.06), wavelet_mean1_ch4 (≈ 0.05), corr_ch0_ch10 (≈ 0.04), and corr_ch4_ch18 / corr_ch3_ch7 (≈ 0.03). Beta-band features (beta_ch18, beta_ch13) and correlation-based connectivity measures dominated the top-ranked features, underscoring the critical role of fronto-central and occipital synchronization in ADHD detection. Physiologically, corr_ch4_ch6 represents fronto-central connectivity (C4-F6), implicating impaired inhibitory control and attentional modulation both hallmark deficits in ADHD. The feature wavelet_mean1_ch4 highlights transient time-frequency irregularities in C4 related to sensorimotor regulation. Collectively, these findings demonstrate that the ensemble classifier effectively captures distributed neural dysregulation through frequency-domain and connectivity-based biomarkers, reinforcing the interpretability and neurobiological relevance of the MRS-SHAP-selected features.

3.9 Model validation, statistical robustness, and diagnostic visualization

The predictive power of the proposed model was further strengthened through a stratified nested cross-validation strategy combined with SMOTEENN resampling, effectively

mitigating class imbalance and demographic bias. This rigorous validation framework ensures that the classification results are statistically robust, generalizable, and reproducible across diverse EEG populations. The integration of SHAP-based interpretability not only enhances model transparency but also guarantees that the most influential features correspond to neurophysiologically validated markers of ADHD. The selected feature set dominated by beta and theta spectral components, microstate stability metrics, and inter-regional correlation features aligns closely with established evidence linking ADHD to frontal hypoactivation, temporal instability, and functional connectivity disruptions. Hence, the model successfully bridges predictive performance and biological plausibility, advancing explainable EEG-based ADHD diagnostics. Model evaluation outcomes are illustrated in Figs. 12 and 13. Figure 14 presents the confusion matrix for the ensemble Voting Classifier on the test set, demonstrating strong classification capability with 541 true negatives (correctly identified controls) and 701 true positives (accurately classified ADHD cases), along with only 48 false positives and 52 false negatives.

This high concordance between predicted and actual labels indicates strong classification performance within the evaluated dataset, with a favorable balance between sensitivity and specificity. The Receiver Operating Characteristic (ROC) curve as shown in Fig. 15 demonstrates a high Area Under the Curve ($AUC = 0.98$), reflecting excellent discriminative capability under the current experimental conditions. The proximity of the ROC curve to the top-left corner further supports the model's ability to effectively distinguish between ADHD and control groups across varying decision thresholds. Despite these encouraging results, the findings should be interpreted within a methodological context. The evaluation is limited to a single dataset and does not include validation on independent cohorts or real-world clinical settings. Therefore, the reported performance does not establish clinical reliability or readiness for deployment. Overall, the proposed MRS-SHAP framework demonstrates promising predictive performance

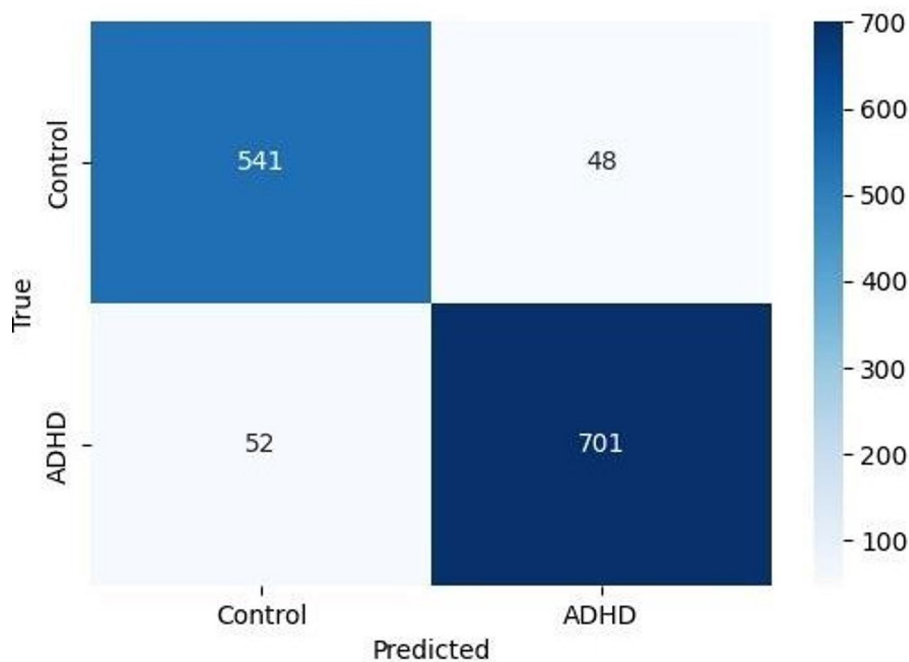


Fig. 14 Confusion matrix showing classification performance of Voting Classifier model

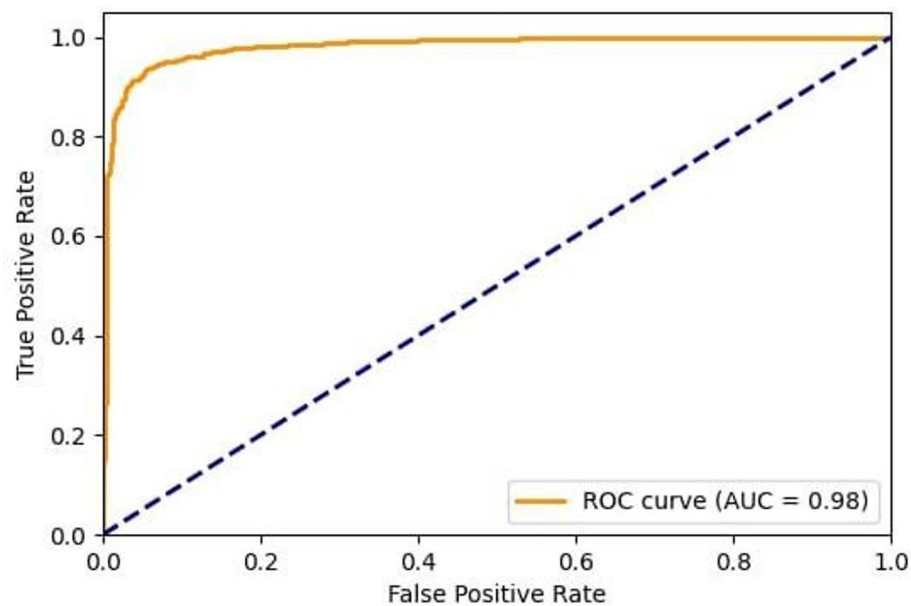


Fig. 15 ROC-AUC curve for Voting Classifier

Table 11 Top ten EEG features selected by the MRS-SHAP framework

Rank	Feature Name	Type
1	beta_ch18	Channel Feature
2	gfp_mean	Microstate
3	microstate_duration_0	Microstate
4	corr_ch3_ch7	Correlation
5	corr_ch4_ch6	Correlation
6	microstate_duration_3	Microstate
7	microstate_duration_1	Microstate
8	corr_ch2_ch10	Correlation
9	corr_ch4_ch8	Correlation
10	microstate_duration_2	Microstate

and interpretability for EEG-based ADHD classification. However, further validation through multi-center studies, external datasets, and prospective clinical investigations is required to determine its generalizability and potential clinical utility.

3.10 Neurophysiological interpretation and clinical implications of top-ranked features

The final selected feature set comprised ten EEG-derived features spanning three key domains: spectral (channel-specific), microstate-based, and connectivity (correlation-based), which collectively represent the most discriminative and neurophysiologically relevant biomarkers for ADHD classification. Table 11 summarizes this ranked feature list.

The highest-ranked feature, *beta_ch18*, reflects beta-band (13–22 Hz) power at channel 18, typically localized over the posterior parietal or occipital cortex. Beta oscillations are associated with sustained attention, cognitive control, and motor regulation functions frequently disrupted in ADHD. This finding aligns with prior reports of reduced beta synchronization and sensory coherence in ADHD populations, indicating deficits in visual attention modulation. Importantly, the identification of this feature through SHAP-based attribution reinforces the utility of explainable artificial intelligence (XAI)

in uncovering neurophysiologically relevant biomarkers, as SHAP enables model-agnostic interpretation of feature contributions at both local and global levels [69]. The second-ranked feature, *gfp_mean*, quantifies Global Field Power (GFP), representing the standard deviation of EEG potentials across electrodes and capturing whole-brain synchronization strength. Elevated GFP may reflect increased cortical excitability and compensatory synchronization, both observed in ADHD, particularly during cognitively demanding tasks. From an XAI perspective, the consistent prioritization of such global descriptors highlights the ability of SHAP to capture both localized and distributed neural dynamics, which are often overlooked in conventional feature selection approaches. Microstate duration features (*microstate_duration_0–3*) capture temporal instability in global brain networks. Specifically:

- *microstate_duration_0* corresponds to the auditory network,
- *microstate_duration_1* reflects the visual network,
- *microstate_duration_2* represents the default mode network (DMN), and,
- *microstate_duration_3* corresponds to the dorsal attention network (DAN).

Reduced microstate durations observed here suggest impaired network persistence and excessive switching, consistent with ADHD's hallmark temporal dysregulation. These findings reinforce and extend current DMN-centric models by emphasizing dynamic network instability as a biomarker of ADHD. Correlation-based connectivity features further highlight ADHD's distributed neural dysfunctions:

- *corr_ch3_ch7* (F3-T7) represents left prefrontal-temporal connectivity, associated with verbal working memory.
- *corr_ch4_ch6* (Fz-F8) indicates fronto-frontal inhibitory control.
- *corr_ch2_ch10* (F7-C4) bridges language and motor integration pathways.
- *corr_ch4_ch8* (Fz-O2) reflects top-down visual attention modulation.

This combination of spatially and functionally diverse correlations supports the view that ADHD is associated with distributed connectivity alterations rather than isolated cortical deficits, consistent with findings from functional neuroimaging studies. The identified features, including beta-band activity, microstate dynamics, and connectivity measures, demonstrate strong neurophysiological relevance and contribute to the interpretability of the proposed model. However, their potential clinical implications should be interpreted with caution. While prior studies have linked posterior beta activity to treatment response and microstate patterns to network stability, the present study does not directly evaluate treatment outcomes or longitudinal effects. Therefore, any inference regarding treatment monitoring, patient stratification, or neuromodulatory targeting remains speculative and beyond the scope of the current analysis. Overall, the physiological interpretability and robustness of these features support the effectiveness of the MRS-SHAP framework for EEG-based ADHD classification within a research setting.

Nevertheless, the proposed model should be regarded as a methodological contribution rather than a clinically validated diagnostic or decision-support system. Further validation across independent cohorts, longitudinal studies, and clinically controlled settings is required to establish its translational relevance and broader applicability. Explainable artificial intelligence (XAI) techniques, particularly SHapley Additive exPlanations

(SHAP), have been widely adopted for interpreting complex machine learning models in biomedical applications. Recent studies demonstrate that SHAP can effectively identify spatial and temporal EEG components contributing to classification decisions, enabling objective biomarker discovery [69]. Furthermore, integrating SHAP with ensemble models such as XGBoost enhances model transparency and supports clinical decision-making by providing explainable and actionable insights [70]. The growing adoption of SHAP across biomedical domains underscores its robustness for feature importance estimation and model interpretability.

Notably, subject-level cross-validation yielded lower performance compared to segment-level evaluation, providing a more realistic estimate of generalization under subject-wise data separation and highlighting the importance of strict validation protocols in EEG-based modeling. It is also important to consider potential confounding effects related to medication exposure. The observed spectral and connectivity differences, including elevated beta-band activity, may reflect the influence of stimulant medication rather than purely disorder-specific neural signatures. Consequently, these findings may represent cohort-specific characteristics, emphasizing the need to disentangle pharmacological effects from underlying neurodevelopmental mechanisms in future EEG-based ADHD studies.

3.11 Subject-level cross-validation using GroupKFold

To ensure a rigorous and leakage-free evaluation, subject-level cross-validation was performed using the GroupKFold strategy. In this approach, all EEG segments corresponding to a single subject were assigned exclusively to either the training or testing set within each fold, preventing overlap of subject-specific data across splits. This evaluation contrasts with segment-level cross-validation, where EEG recordings are divided into multiple windows and randomly split, potentially allowing segments from the same subject to appear in both training and testing sets. While segment-level evaluation often yields higher performance due to intra-subject similarity, it may overestimate generalization capability. Under subject-level cross-validation, the proposed model achieved an average accuracy of 79.50%, F1-score of 81.06%, and AUC of 0.8650, representing a more realistic estimate of performance on unseen subjects as shown in Table 12. Additionally, evaluation on a held-out test set further reduced performance (Accuracy ≈ 65.72%, AUC ≈ 0.73), held-out test set consisted of approximately 20% of subjects (~ 24 participants) with balanced representation of ADHD and control groups. Model evaluation was performed at the segment level, where each subject contributed multiple EEG segments as shown in Table 13 and highlighting the inherent challenges of generalizing EEG-based models. These results emphasize the importance of subject-level validation in EEG classification tasks and provide a transparent assessment of the proposed framework’s generalization capability.

Table 12 Subject-Level Cross-Validation Performance Using GroupKFold (5-Fold)

Fold	Sensitivity	Specificity	F1-Score	Accuracy	AUC
1	0.7472	0.8565	0.8040	0.7950	0.8797
2	0.8150	0.6904	0.8409	0.7795	0.8299
3	0.8276	0.6261	0.6799	0.7029	0.8451
4	0.8379	0.7990	0.8379	0.8205	0.8543
5	0.8759	0.8791	0.8901	0.8773	0.9160
Mean ± SD	0.8207 ± 0.05	0.7702 ± 0.11	0.8106 ± 0.08	0.7950 ± 0.06	0.8650 ± 0.03

Table 13 Held-out Test Set Results

Metric	CV	Held-out
Accuracy	79.5%	65.7%
AUC	0.865	0.733

Table 14 Impact of SMOTE-ENN on Model Performance (Subject-Level Split)

Setting	Accuracy	F1-Score
Without SMOTE-ENN	0.9042	0.9192
With SMOTE-ENN	0.8954	0.9104

Table 15 Ablation analysis of the proposed MRS-SHAP framework

Method	Accuracy	F1-Score
SHAP	0.9048	0.9197
DTW	0.9025	0.9176
SHAP + DTW	0.9119	0.9260
SHAP + BW	0.9048	0.9197
MRS (Proposed)	0.9101	0.9243

This evaluation directly addresses potential data leakage concerns and ensures that the reported performance is not influenced by intra-subject similarities. The high AUC (≈ 0.98) reported under segment-level evaluation is therefore interpreted as an optimistic estimate, while subject-level and held-out results provide a more clinically realistic performance.

3.12 Ablation and sensitivity analysis

To evaluate the contribution of individual components and the robustness of the proposed framework, a series of ablation and sensitivity analyses were conducted under leakage-free subject-level splitting.

As shown in Table 14, the inclusion of SMOTE-ENN resulted in only marginal changes in performance (Accuracy: 0.9042 $\rightarrow$ 0.8954; F1: 0.9192 $\rightarrow$ 0.9104). This indicates that the proposed approach does not rely on synthetic oversampling, and that performance improvements are primarily driven by feature selection and model design rather than data augmentation.

The contribution of each component within the MRS-SHAP framework is summarized in Table 15. SHAP-only and DTW-only approaches achieved comparable performance (Accuracy ≈ 0.90), indicating that both feature importance and temporal stability independently capture relevant EEG characteristics. However, combining SHAP and DTW resulted in improved performance (Accuracy = 0.9119, F1 = 0.9260), demonstrating their complementary roles. The SHAP + BW configuration did not provide additional gains over SHAP alone, suggesting that performance-based weighting alone is insufficient without temporal stability. The full MRS formulation achieved consistent performance (Accuracy = 0.9101, F1 = 0.9243), confirming that the proposed multi-objective integration effectively balances feature importance and temporal consistency.

Table 16 Feature retention sensitivity analysis

Retention (%)	Accuracy	F1-Score
30%	0.9019	0.9168
40%	0.9036	0.9190
50%	0.9113	0.9256
60%	0.9101	0.9243
70%	0.9031	0.9180

To assess robustness, the feature retention threshold was varied from 30% to 70% as shown Table 16. Performance improved from 30% to 50%, reaching optimal values at 50% (Accuracy = 0.9113, F1 = 0.9256), and declined slightly thereafter, indicating that excessive feature inclusion introduces redundancy. The selected threshold of 60% lies within this optimal range, providing a balance between accuracy and feature diversity. These results demonstrate that the framework is stable across moderate variations in feature selection parameters. Collectively, these analyses confirm that the proposed MRS-SHAP framework derives its performance from the synergistic integration of feature importance and temporal stability, rather than reliance on resampling or highly sensitive parameter tuning

4 Conclusion

This study introduces a Multi-Objective Rank Scored SHapley Additive exPlanations (MRS-SHAP) framework for EEG-based ADHD classification, integrating SHAP-derived feature importance, DTW-based temporal stability, and performance-aware weighting within a unified feature selection strategy. This approach enables the identification of compact, robust, and neurophysiologically relevant EEG biomarkers. The framework achieved high performance under segment-level evaluation (Accuracy ≈ 98.40%, F1-score ≈ 98.40%), representing an upper-bound estimate. Under stricter and clinically relevant validation, subject-level cross-validation yielded moderate performance (Accuracy ≈ 79.5%, AUC ≈ 0.86), while evaluation on a held-out test set (~ 20% of subjects) produced a more conservative estimate (Accuracy ≈ 65.7%, AUC ≈ 0.73). This performance gap highlights the impact of inter-subject variability and underscores the necessity of subject-level validation in EEG-based modeling.

The selected features primarily beta-band activity, microstate dynamics, and inter-regional connectivity across frontal, central, and occipital regions—are consistent with established neurophysiological evidence of ADHD-related network dysfunction. SHAP-based analysis further confirms that model decisions are driven by biologically relevant patterns, enhancing interpretability and methodological validity. Ablation and sensitivity analyses demonstrate that integrating feature importance with temporal stability improves robustness, with minimal dependence on synthetic resampling. These findings validate the effectiveness of the proposed multi-objective feature selection strategy. Despite these strengths, the study is limited to a single dataset and lacks external validation across independent cohorts. Additionally, potential confounding effects, such as medication exposure, may influence observed EEG patterns. Accordingly, the findings should be interpreted as methodological validation rather than clinical evidence.

Future work will focus on cross-dataset and multi-center validation, as well as the integration of longitudinal and clinically controlled data to assess generalizability and translational relevance. Overall, the MRS-SHAP framework provides a robust and interpretable approach for EEG feature selection, representing a methodological advancement toward explainable and physiologically grounded ADHD analysis, though not yet a clinically deployable diagnostic tool.

4.1 Limitations and future work

Despite the promising performance and interpretability of the proposed MRS-SHAP framework, several limitations should be acknowledged. First, the study is based on a

single-site dataset comprising 121 pediatric subjects, which may limit generalizability across different populations, recording conditions, and acquisition systems. Although subject-level cross-validation was employed to mitigate optimistic bias, external validation across independent datasets and multi-center cohorts remains essential to establish robustness. Second, while segment-level evaluation demonstrated high performance, it represents an upper-bound estimate due to potential intra-subject similarity. More realistic evaluation through subject-level cross-validation and held-out testing revealed moderate generalization performance, highlighting the inherent variability of EEG signals across individuals. Third, potential confounding effects related to medication exposure were not controlled. The ADHD cohort included subjects treated with methylphenidate, which is known to influence EEG dynamics, particularly beta-band activity and functional connectivity. Due to the absence of detailed medication metadata, it was not possible to isolate disorder-specific neural signatures.

Fourth, the study focuses on classical machine learning models, and comparisons with recent advanced EEG model architectures, such as convolutional neural networks, transformers, or graph-based models, were not performed. While this choice supports interpretability, it limits direct comparison with modern approaches. Fifth, preprocessing and feature selection choices including weighted ICA, notch filtering, DTW-based temporal stability, and feature retention thresholds were empirically defined. Although ablation and sensitivity analyses were conducted, a more exhaustive evaluation of preprocessing strategies and parameter configurations could further improve reproducibility and standardization. Sixth, the absence of detailed clinical severity scores prevents correlation between EEG-derived features and symptom-level manifestations of ADHD, limiting the clinical interpretability of the identified biomarkers. Finally, computational performance metrics such as runtime, memory usage, and inference latency were not systematically evaluated, which are important considerations for real-world deployment.

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Author contributions

Mohammed Atheef G A: Original Draft, Software, Methodology, Conceptualization, Data Visualization, Review and Editing. Shyam Pranav G: Resources, Original Draft, Software, Methodology, Conceptualization, Data Visualization. Omkar S Powar: Supervision, Methodology, Writing: Review and Editing. All authors read and approved the final version of the manuscript.

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Data availability

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Declarations

Ethical approval

This study utilized a publicly available, de-identified EEG dataset. The original data collection was conducted in accordance with relevant ethical guidelines and approved by the appropriate institutional review board. Therefore, no additional ethical approval was required for this secondary analysis.

Consent to participate

Not applicable. The study is based on a publicly available, anonymized dataset, and informed consent was obtained by the original data providers.

Consent to publish

Not applicable. The manuscript does not contain any individual person's data in any identifiable form.

Competing interests

The authors declare no competing interests.

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